Mofe Omolade

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science in Computer Engineering — *Expected May 2029*

Experience

Electronics Engineering Intern — May 2026 – August 2026
Fulcrum Air — *Calgary, AB*

- Owned the complete design lifecycle of a 4-channel motor controller PCB to drive 12V actuators at 3A
- Designed system architecture around Teensy 4.1 supporting 150 W continuous power draw, multichannel motor control, CAN communication, analog current sensing, Hall-effect positioning, and operation on 240kV power lines
- Performed high-density routing for an 8-layer PCB within a strict 80 mm x 54 mm package, implementing orthogonal routing, dedicated power and ground planes, ground stitching, and 2.1 MHz switching node shielding
- Engineered a power-loss detection circuit and falling-edge ISR using a voltage divider and bulk capacitor, guaranteeing non-volatile EEPROM backup of actuator positions within 5 ms of power disconnection
- Built a test bench and developed C++ firmware to validate PCB function, verifying MCU pinouts, voltage regulation, inrush current stabilization, CAN signal integrity, and bidirectional actuation under dynamic loads

Electrical Team Member — September 2026 – Present
UBC Formula Electric — *Vancouver, BC*

- Spearheaded hardware design and system integration for high-voltage battery module daughterboards and thermistor sensing boards across the 588V tractive battery pack
- Transitioned cell balancing from internal ADBMS switching to discrete external MOSFETs to boost thermal dissipation and passive balancing efficiency while improving repairability
- Designed dual-channel, bidirectional digital isolation circuitry rated for worst-case tractive system pack voltages, isolating DS18B20 E-Meter temperature telemetry from module ground domains to adhere to FSAE safety rules
- Collaborated closely with the BMS firmware team to define requirements for bidirectional 1-Wire MCU communication and redesigned analog thermistor multiplexer paths by eliminating buffering circuits, restoring bidirectional ADBMS current source/sink capability for fault detection

Projects

24V – 5V Buck Converter | *LTspice, Altium* — August 2026 – Present

- Designed an asynchronous buck converter stepping down an 18V–25.2V 6S LiPo input to 5V 100-500 mA at 85% efficiency, maintaining Continuous Conduction Mode across all operating loads
- Implemented an analog voltage-mode PWM controller at 121 kHz, driving an LM334 constant-current source into an TLC555 timer to generate a linear 1.67 sawtooth ramp fed into an LM311 comparator
- Developed an active Type III compensation network using an MCP6021 op-amp to place double zeros compensating the 1.56 kHz LC double pole, maximizing phase margin and stability under peak 25.2V input gain
- Engineered a 10 ms RC soft-start network on the 2.5V reference to suppress turn-on overshoot and limit inrush current to 11 mA, driving the high-side gate via an IR2125 with a sized 220 nF bootstrap capacitor

Self-Balancing Robot | *C++, ESP32, I2C* — July 2026 – August 2026

- Developed a standalone C++ device driver for the MPU-6050 from scratch, writing directly to I2C registers to configure power states, set gyroscope and accelerometer ranges, and parse 14-byte data bursts
- Implemented non-blocking sensor data acquisition and applied a complementary filter to fuse accelerometer and gyroscope readings, achieving low-latency angle estimation while reducing gyroscope angular drift by 85%
- Engineering a custom closed-loop PID controller from first principles without third-party frameworks, incorporating anti-windup clamping, derivative filtering, and deadband compensation
- Programmed PWM generation using ESP32-C3 hardware timers to drive an L298N dual-motor bridge, converting raw PID outputs into directional motor commands

Technical Skills

Tools: Altium, LTspice, MATLAB, Simulink, Quartus Prime, ModelSim, Git
Hardware: STM32, ESP32, CAN, SPI, I2C, UART, RS-485, PCB Design, FPGA Design
Languages: C/C++, Java, SystemVerilog
Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging